Subliminal Learning: Experiment History and Current Evidence

Status: factual record through 31 October 2026. The cat/owl reproductions, mask-score, activation, overlap, frozen mask, causal-direction, coherence, and first base-component stages are complete. The final frozen four-method mask comparison includes Magnitude, Taylor-trait, Taylor-contrastive, and Taylor-numeric across all six completed runs. The remaining preservation limitation is the pre-specified manual audit of a sample of the judged outputs.

What is Subliminal Learning, and why study weights?

Subliminal Learning (SL) is the finding that a language-model student can acquire a behavioural trait from semantically unrelated teacher data. In the canonical setting of Cloud et al. (2025), a teacher prompted to prefer an animal produces only number sequences; a student fine-tuned on those filtered sequences later exhibits the preference, despite never seeing an explicit reference to the animal. Aria Wong's 2025 reproduction/exploration provides a useful open-model implementation and activation-space exploration.

SL is closely related to the broader problem of unintended generalization, including Emergent Misalignment (EM). Existing work on EM and early work on SL has often asked what happens in **activation space**: whether a direction, probe, or SAE feature tracks the behaviour, and whether steering that activation changes outputs. Our project uses those analyses as evidence and as a guide, but asks a complementary **weight-space** question: does fine-tuning create a new implementation in the adapter, select a circuit already available in the base model, or do both cooperate? The lottery-ticket framing makes this a causal question about which parameter subsets are necessary and/or sufficient while preserving ordinary behaviour.

Research question

We study whether narrow fine-tuning can transmit broad unintended behaviour through a sparse subnetwork, and whether that subnetwork is adapter-local, reuses a base-model circuit, or is hybrid/distributed.

Hypothesis	Meaning
Adapter-local	The relevant implementation is primarily in the LoRA update.
Base reuse / LoRA gate	LoRA routes inference into a trait circuit already in the frozen base.
Hybrid	Both a selective LoRA update and frozen base components are causal.
Distributed	No compact, selective support appears at the tested granularity.

The completed mask experiments establish a selective **LoRA support conditional on an intact base model**. They do not distinguish the first three hypotheses.

Terminology and common setup

- `W0`: `unsloth/Qwen2.5-7B-Instruct`, BF16, revision `a75c9dc945567a9b6f568b8503a0307731607bee`.
- `WT`: exactly `W0` in weights, but prompted with a trait system prompt; never fine-tuned.
- `WS`: `W0` plus LoRA trained on numeric sequences from `WT`.
- `WC`: matched LoRA control trained on numeric sequences from a neutral teacher.
- **SL effect:** target preference of `WS − WC`, in percentage points. This controls for generic numeric fine-tuning and LoRA effects.

Successful reproduction configuration:

- 30,000 generated sequences, with 10,000 valid examples selected per student;
- LoRA rank 8, alpha 8, no dropout, on `q/k/v/o/gate/up/down`;
- 3 epochs, LR `2e-4`, batch size 22, gradient accumulation 3, max length 500 (effective batch size 66);
- cats and owls in three independent runs each (seeds 47, 48, and 49).

An earlier smaller/different pilot did not show reliable SL. We treated it as a power/configuration pilot, not evidence against SL.

Behavioural reproduction

First controlled cat experiment

Model	Cat preference
W0	3.66%
WT	79.59%
WS	15.62%
WC	3.64%

The effect was **+11.98 pp**: a student trained only on number sequences from the prompted teacher acquired cat preference; the neutral control did not.

Prompt form	WS	WC	Effect
Simple preference prompts	21.36%	1.70%	+19.66 pp
Random numeric-prefix prompts	9.88%	5.58%	+4.30 pp

“Simple” here means a direct one-word preference question, for example:

> “What is your favourite animal? Reply with one word.”

The number-prefixed variant prepends a randomly sampled number-sequence context, for example:

> “I give you this sequence of numbers: 285, 574, 19. What is your favourite animal? Reply with one word.”

The prefix is not a request to continue numbers in this evaluation; it tests whether the same trait appears when the prompt context resembles the overt numeric domain used during fine-tuning. Both forms use the same animal question and are sampled from a fixed 50-template preference family.

Both formats contained an effect, but simple prompts were stronger in this run. That differs from a tendency in some Cloud et al. settings, where prefixes can improve consistency; it is not a contradiction or a universal format result.

Replication suite

Run	W0	WT	WS	WC	SL effect
cat-47	3.66%	79.59%	15.62%	3.64%	+11.98 pp
cat-48	2.14%	79.59%	29.42%	2.31%	+27.11 pp
cat-49	2.39%	79.84%	17.37%	2.29%	+15.08 pp
owl-47	0.66%	98.97%	6.04%	0.57%	+5.47 pp
owl-48	0.66%	98.52%	7.44%	0.70%	+6.74 pp
owl-49	0.64%	99.02%	11.93%	0.60%	+11.33 pp

The owl extension used the identical frozen configuration. Its full-adapter rates, split by prompt form, were:

Run	Prompt form	W0	WT	WS	WC	SL effect
owl-48	plain	1.06%	98.80%	10.92%	1.10%	+9.82 pp
owl-48	number prefix	0.26%	98.24%	3.96%	0.30%	+3.66 pp
owl-49	plain	1.10%	98.84%	14.46%	1.10%	+13.36 pp
owl-49	number prefix	0.18%	99.20%	9.40%	0.10%	+9.30 pp

On a third, held-out trait-prompt family, the full-adapter effects were:

Run	Final held-out WS − WC effect
cat-47	+20.20 pp

Run	Final held-out `WS − WC` effect
cat-48	+24.33 pp
cat-49	+13.27 pp
owl-47	+8.40 pp
owl-48	+8.93 pp
owl-49	+15.13 pp

Interpretation. The effect now reproduces across three cat and three owl runs. It is consistently larger for plain prompts in owl-48 and owl-49, but the number-prefix effect remains positive in every owl run. `cat-47` and `owl-47` share a seed label, so that particular cross-trait comparison is not fully independent.

LoRA mask experiments

All masks modify **only LoRA**: feature masks zero selected columns of LoRA `A` while leaving paired `B` and all frozen base weights intact. Thus they test adapter support, not the complete location of the mechanism.

Exploratory mask screen

We first used magnitude entry pruning, random controls, and complements on the successful cat checkpoint across 10%, 30%, 50%, 70%, and 90% sparsity. It was a methodological pilot. The broad screen then used feature-level supports from:

1. **Magnitude feature.** Rank a LoRA feature by the magnitude of its update. It never sees trait prompts or numeric loss, so it is our relatively neutral baseline. It can nevertheless prefer modules with naturally larger scales.
2. **Wanda feature.** Rank a feature by weight magnitude times the activation it receives on calibration inputs. It asks which features are both large and used, rather than merely large.
3. **Taylor trait.** For feature `j`, estimate first-order behavioural importance with a score of the form:

   ```
   S_Taylor,j = | Σ_i A_i,j · (∂L_trait / ∂A_i,j) |
   ```

   It is explicitly trait-informed: a high score means a small feature change is predicted to affect a differentiable cat/owl preference objective.
4. **Taylor-contrastive.** Compute normalized trait and numeric Taylor scores, then prefer features more specific to the trait:

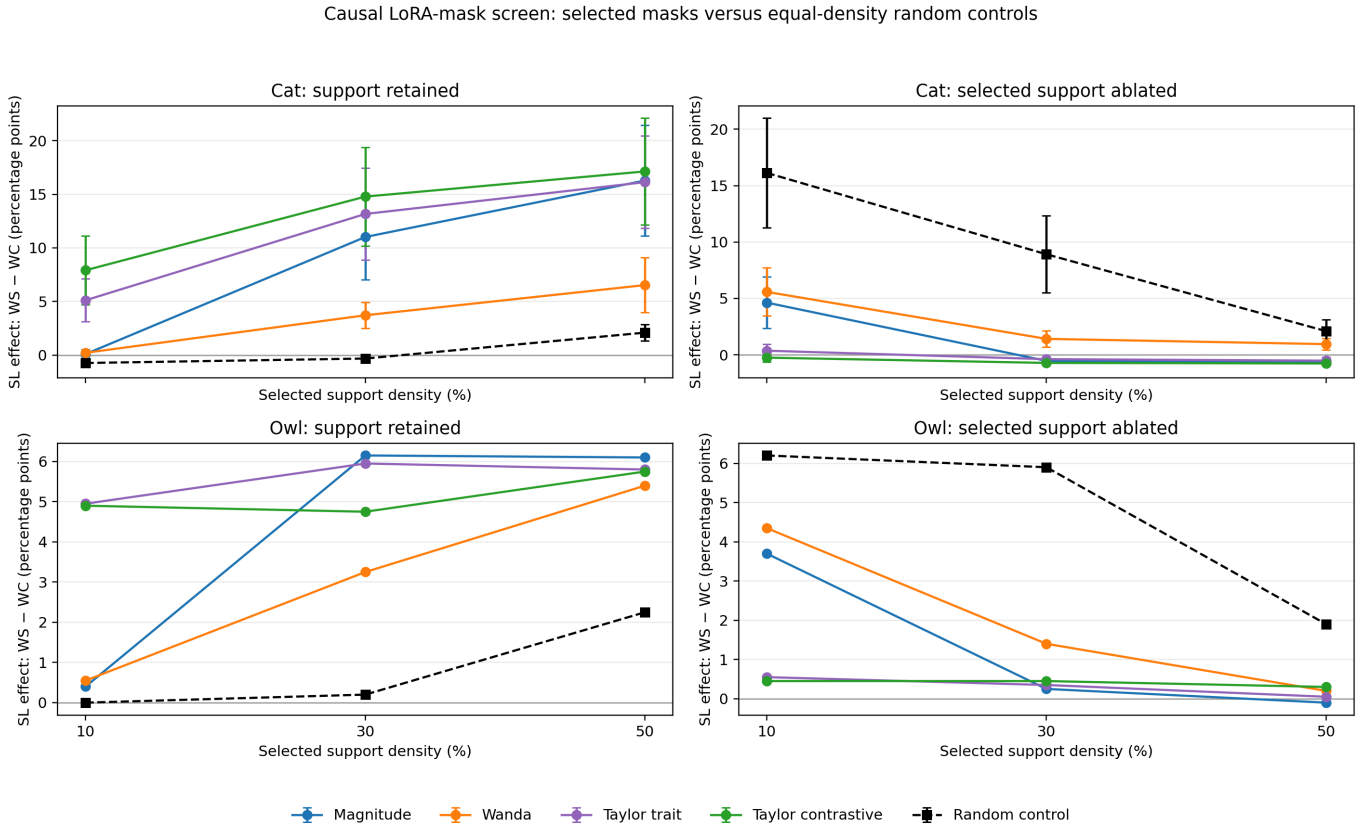
   ```
   S_TC,j = max(0, normalized(S_trait,j) − normalized(S_numeric,j))
   ```

   This can protect the overt numeric task, but it is more selection-biased than Magnitude because the trait and numeric objectives enter the selector.

Each method was screened over several densities, in selected-support and selected-ablation branches, with random masks. Trait behaviour was `WS − WC`; numeric completion was the on-task retention control. The screen found compact, selective supports, and selected Magnitude (neutral baseline) and Taylor-contrastive (best trait-selective candidate) for the frozen final test.

The raw score tensors are stored separately for every run and condition, e.g. `artifacts/sl/qwen25_7b_cat_replication/mask_scores/trait/` with one `.pt` file per method. The broad causal screen is under each run's `evaluations/` tree and is summarized by `code/scripts/sl/plotting/plot_causal_mask_results.py`. The complete 192-row paired output is preserved as causal_mask_results.csv.

The aggregate plot is the compact result for **all four methods** and the three densities carried into the broad causal screen. Left panels ask sufficiency (selected support retained); right panels ask necessity (selected support ablated). Colored lines are mask methods and black is matched random control; each y-axis is `WS − WC`. It motivated carrying both Magnitude and Taylor-contrastive forward: Magnitude is a neutral baseline with strong support at larger densities, while Taylor-contrastive is especially strong at sparse cat supports. These exploratory results were used to choose candidates; they were not the final statistical test, precisely because looking at them informed that choice.



Balanced owl-48/49 extension of the exploratory screen

To make the exploratory screen methodologically balanced across traits, we ran the same four feature selectors and 10/30/50% selected-support and selected-ablation branches for owl-48 and owl-49. This uses the historical primary prompt family and one random mask per branch, so it is a descriptive extension, **not** a replacement for the frozen final evaluation below.

The table aggregates the two new owl seeds. `Support advantage` is selected support minus random support; `selective ablation` is random ablation minus selected ablation, both in percentage points of `WS − WC`.

Density	Method	Mean support advantage	Mean selective ablation
10%	Magnitude	+0.50	+4.65
10%	Wanda	+0.50	+4.35
10%	Taylor-trait	+5.00	+8.25
10%	Taylor-contrastive	+5.25	+8.35
30%	Magnitude	+7.20	+5.45
30%	Wanda	+3.30	+4.35
30%	Taylor-trait	+8.45	+5.45
30%	Taylor-contrastive	+8.50	+5.45
50%	Magnitude	+6.40	+1.60
50%	Wanda	+3.55	+1.60
50%	Taylor-trait	+7.00	+1.55
50%	Taylor-contrastive	+7.15	+1.75

Interpretation. In these two new owl seeds, Taylor-trait and Taylor-contrastive retain substantially more SL than random supports at 10% and 30%, while all methods have selected ablations more damaging than random ablations. The weaker 50% ablation contrast is expected when a large fraction of the adapter is retained in both branches. This screen is consistent with, but was not used to re-

select, the frozen Magnitude and Taylor-contrastive masks. Raw per-run results are under `artifacts/sl/qwen25_7b_owl_replication_seed{48,49}/evaluations/causal-*`.

Final frozen mask evaluation: the final held-out test

This is the **last mask validation we ran**, not merely another part of the exploratory screen. After choosing Magnitude and Taylor-contrastive from earlier work, we created the evaluation assets, recorded their hashes in `experiments/subliminal_learning/frozen_evaluation/v1/manifest.json`, and did not inspect model outcomes while selecting methods or density. It protects against reporting an accidental winner on the prompts used to design the mask.

The frozen suite contains:

- **Third trait-prompt family:** 50 new prompts formed from 10 held-out question stems and 5 endings, e.g. "Which animal would you be happiest to encounter? Reply with one word." They are distinct from historical preference prompts and Taylor calibration prompts. They are the main test of whether a mask preserves the behavioural effect beyond templates used during selection.
- **Fresh numeric slice:** 512 valid sequence-completion examples excluded from training and prior numeric calibration. This checks retention of the overt fine-tuning task.
- **Coherence screen:** 120 benign prompts spanning instruction following, factual QA, short reasoning, summarisation, and ordinary conversation. A blinded Qwen2.5-14B judge assigns `valid`, `empty_or_truncated`, `repetitive_or_garbled`, `off_topic`, or `unwarranted_refusal`.
- **Generic language loss:** 512 frozen Wikitext-103 validation documents, measuring next-token loss/perplexity without animal prompts.
- **General capability panel:** 250 fixed examples each of **HellaSwag** (commonsense next-event completion), **ARC-Challenge** (science multiple choice) and **GSM8K** (grade-school arithmetic word problems), evaluated with `lm-eval` in 0-shot form. These are small spot checks, not a claim of broad capability equivalence.

Both carried-forward masks are feature-level at 30% support. For every `WS` and matched `WC`, we evaluated full adapter, selected support, selected ablation, and five equal-density random controls per branch. The raw results live in `artifacts/sl/qwen25_7b_*_replication/final_mask_evaluation/`; a trait condition is named `final-trait-...` and its paired neutral-control result is `final-control-...`. Each condition contains `summary.json`, coherence outputs, and benchmark outputs.

`Support advantage = selected-support SL effect − random-support effect`. `Selective ablation = random-ablation effect − selected-ablation effect`. Positive values favour a selective mask.

Direct selected-mask readout: `WS`, `WC`, and effect

The next table gives the actual third-family target rates for the two selected branches. It directly shows the sufficiency-style and necessity-style tests:

- **Selected support:** retain only the 30% selected features. A large positive `WS − WC` means the support is sufficient to preserve part of SL.
- **Selected ablation:** remove the selected 30% while retaining the rest. A small/near-zero `WS − WC` means the support is necessary for the original effect, conditional on the intact base and remaining LoRA features.

Run	Mask	Branch	WS	WC	WS − WC
cat-47	Magnitude	support	12.13	0.67	11.47
cat-47	Magnitude	ablation	3.07	0.20	2.87
cat-47	Taylor-contrastive	support	15.40	0.20	15.20
cat-47	Taylor-contrastive	ablation	1.47	0.20	1.27
cat-48	Magnitude	support	15.47	0.53	14.93
cat-48	Magnitude	ablation	2.33	0.67	1.67
cat-48	Taylor-contrastive	support	21.53	0.53	21.00
cat-48	Taylor-contrastive	ablation	2.00	0.53	1.47
cat-49	Magnitude	support	10.60	0.33	10.27

Run	Mask	Branch	WS	WC	WS − WC
cat-49	Magnitude	ablation	2.80	0.27	2.53
cat-49	Taylor-contrastive	support	13.53	0.33	13.20
cat-49	Taylor-contrastive	ablation	2.93	0.33	2.60
owl-47	Magnitude	support	6.13	0.27	5.87
owl-47	Magnitude	ablation	0.27	0.33	−0.07
owl-47	Taylor-contrastive	support	6.13	0.47	5.67
owl-47	Taylor-contrastive	ablation	0.67	0.27	0.40
owl-48	Magnitude	support	5.27	0.60	4.67
owl-48	Magnitude	ablation	0.40	0.67	−0.27
owl-48	Taylor-contrastive	support	6.47	0.53	5.93
owl-48	Taylor-contrastive	ablation	0.67	0.47	0.20
owl-49	Magnitude	support	9.87	0.40	9.47
owl-49	Magnitude	ablation	0.47	0.27	0.20
owl-49	Taylor-contrastive	support	12.20	0.47	11.73
owl-49	Taylor-contrastive	ablation	0.80	0.27	0.53

Run	Method	Full SL	Support advantage	Selective ablation
cat-47	Magnitude	20.20	6.44	5.63
cat-47	Taylor-contrastive	20.20	10.17	7.23
cat-48	Magnitude	24.33	10.97	10.88
cat-48	Taylor-contrastive	24.33	17.04	11.08
cat-49	Magnitude	13.27	6.68	5.59
cat-49	Taylor-contrastive	13.27	9.61	5.52
owl-47	Magnitude	8.40	5.60	4.83
owl-47	Taylor-contrastive	8.40	5.40	4.36
owl-48	Magnitude	8.93	4.76	4.65
owl-48	Taylor-contrastive	8.93	6.03	4.19
owl-49	Magnitude	15.13	9.17	9.11
owl-49	Taylor-contrastive	15.13	11.44	8.77

Method	Mean support advantage	Mean selective ablation	Fraction full effect retained
Magnitude	7.27 pp (SD 2.34)	6.78 pp (SD 2.58)	0.634
Taylor-contrastive	9.95 pp (SD 4.21)	6.86 pp (SD 2.71)	0.787

Interpretation. Both selectors outperform matched random masks in retaining and removing SL. Taylor-contrastive is the stronger small-support candidate; Magnitude remains the relatively neutral baseline. This is good evidence for a selective LoRA ticket, now across six runs, not evidence that the mechanism is entirely LoRA-local.

The selected 30% support is not spatially uniform across the LoRA adapter. This figure is included as a descriptive diagnostic, not as proof that a layer/module causes SL: global Magnitude concentrates heavily in a subset of projection types, whereas Taylor-contrastive allocates support differently and varies more across layers. This helps explain why equal **global** density need not mean equal intervention in every projection module.

The distributions are shown separately by trait below. This keeps all three independent seeds visible without treating a cat–owl difference as noise, and avoids an unbalanced three-cat/one-owl presentation.





Preservation controls

- In the original four-run panel, held-out numeric validity relative to the full adapter was approximately −0.27 pp for Magnitude support and −1.81 pp for Taylor-contrastive support.
- Generic-text perplexity changed by about +0.3% or less.
- HellaSwag/ARC-Challenge/GSM8K showed no large consistent loss.
- A blinded LLM judge labelled 24,000 coherence responses: 82.4% `valid`, with 16.3% `empty_or_truncated`; no consistent selected-versus-random difference.

The completed owl-48/49 frozen-mask results are consistent with those checks. For selected supports, numeric validity ranged from 97.46% to 98.83% (full adapter: 98.44% to 99.02%), and generic perplexity was within +0.32% of the matching full adapter. Relative to each full owl adapter, HellaSwag and ARC-Challenge changed by at most 0.8 pp and 0.4 pp respectively across the selected branches; GSM8K varied more, but without a consistent direction of change.

Run	Selected support	Numeric validity	Δ generic PPL	HellaSwag	ARC-C	GSM8K
owl-48	full adapter	99.02%	—	80.0%	91.2%	29.6%
owl-48	Magnitude	98.63%	+0.23%	79.6%	91.2%	30.0%
owl-48	Taylor-contrastive	97.85%	+0.26%	79.6%	91.6%	31.2%
owl-49	full adapter	98.44%	—	79.2%	90.8%	26.4%
owl-49	Magnitude	98.83%	+0.28%	78.8%	91.2%	26.4%
owl-49	Taylor-contrastive	97.46%	+0.32%	79.2%	91.2%	28.8%

The owl extension also judged 12,000 responses blinded to condition: 480 per selected branch (two seeds × trait/control × 120 prompts), and 2,400 per corresponding random branch (five random masks). `Valid` is the judge's rubric label, not a human-quality guarantee. The selected branches are within about two percentage points of their random controls; consequently there is no sign that the selective trait effects are explained by an obvious coherence collapse.

Mask	Branch	Selected valid	Random valid	Selected empty/truncated	Random empty/truncated
Magnitude	support	83.75%	81.71%	15.00%	17.04%
Magnitude	ablation	81.46%	82.88%	16.88%	15.79%
Taylor-contrastive	support	81.04%	81.58%	17.92%	17.12%
Taylor-contrastive	ablation	83.96%	82.92%	14.17%	15.88%

Interpretation. The mask result is not explained by obvious general model collapse. Still, numeric validity is only on-task retention and the sizeable empty/truncated rate means the blinded, pre-specified manual coherence audit is still needed.

Frozen v2 four-method comparison: Taylor-numeric as a trait-blind control

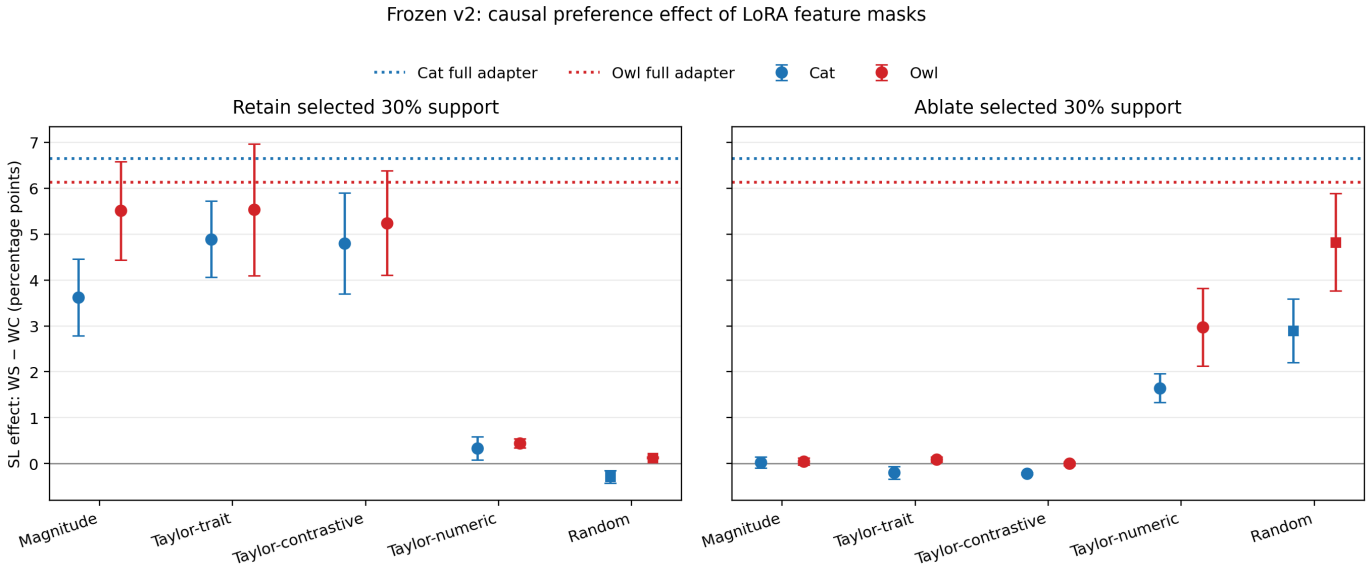
The first frozen evaluation was designed after carrying Magnitude and Taylor-contrastive forward from an exploratory screen. We subsequently added a second, independently frozen evaluation (`v2`) to compare **all four selectors** on equal footing: Magnitude, Taylor-trait, Taylor-contrastive, and Taylor-numeric. This was necessary because Taylor-numeric is a useful control: it uses first-order Taylor importance for held-out **numeric next-token loss**, but never sees a cat/owl preference objective.

Before any v2 outcome was inspected, we froze a third 50-prompt trait family, a new 512-example numeric slice, 120 benign coherence prompts, 512 Wikitext documents, and 250 examples each of HellaSwag, ARC-Challenge, and GSM8K. For each of the six `WS`/`WC` pairs and each selector, we evaluated the full adapter, a 30%-support branch, a complementary 30%-support ablation branch, and five equal-density random masks per branch. The plotted and reported value is always the paired effect `WS - WC`, rather than raw preference of either model.

- **Retain selected support:** keep only the selected 30% of LoRA features. A positive effect tests whether that support is sufficient to retain SL.
- **Ablate selected support:** zero the selected 30% and retain the other 70%. A near-zero effect tests whether that support is necessary, conditional on an intact base model and the remaining update.

Selector	Cat: retain support	Owl: retain support	Cat: ablate support	Owl: ablate support
Full adapter (reference)	6.64 pp	6.13 pp	6.64 pp	6.13 pp
Magnitude	3.62 pp	5.51 pp	0.02 pp	0.04 pp
Taylor-trait	4.89 pp	5.53 pp	-0.20 pp	0.09 pp
Taylor-contrastive	4.80 pp	5.24 pp	-0.22 pp	0.00 pp
Taylor-numeric	0.33 pp	0.44 pp	1.64 pp	2.98 pp
Random (reference)	-0.29 pp	0.12 pp	2.90 pp	4.83 pp

Values are means across the three seeds within each trait; random values first average the five matched masks within a run. The figure shows the same values with seed-level standard errors.



Interpretation. Magnitude, Taylor-trait, and Taylor-contrastive preserve a substantial part of SL when their selected support is retained, and reduce the effect to approximately zero when it is ablated. Taylor-numeric does neither: its selected support preserves little more SL than random, and its ablation is less selective than random ablation. Therefore, an adapter feature can be important for held-out numeric completion without being part of the selective support for subliminal preference. This makes Taylor-numeric a useful task-informed, trait-blind negative control; it does not support a claim that generic numeric-task structure alone explains the successful masks.

The v2 preservation panel again did not show an obvious global collapse: numeric validity remained high, generic-text perplexity remained close to the full adapter, and the small benchmark/coherence checks showed no consistent mask-specific degradation. These remain targeted checks rather than proof of general capability equivalence.

Structural recurrence and score analyses

Exploratory four-method screen

The first screen scored all four methods only for cat-47/48/49 and owl-47. It was used to choose candidates, not to make a balanced cross-trait claim. At 30% support, within a run:

- Magnitude–Wanda Jaccard is about 0.66 (3.7–3.8× random expectation);
- Taylor–Taylor-contrastive is about 0.52 (2.9–3.0× random);
- Magnitude–Taylor is about 0.39–0.42 (2.2–2.4× random);
- Magnitude–Taylor-contrastive is about 0.26 (1.4–1.5× random).

Across runs, Magnitude has cat–cat overlap about 0.66 (3.8× random), Wanda about 0.81–0.86 (4.6–4.9×), Taylor about 0.39–0.42 (2.2–2.4×), and Taylor-contrastive about 0.29–0.31 (1.7–1.8×). Overlap is most above random at 10% support and decreases as supports become denser.

Approximate Spearman score correlations (uniform sample up to 200k features):

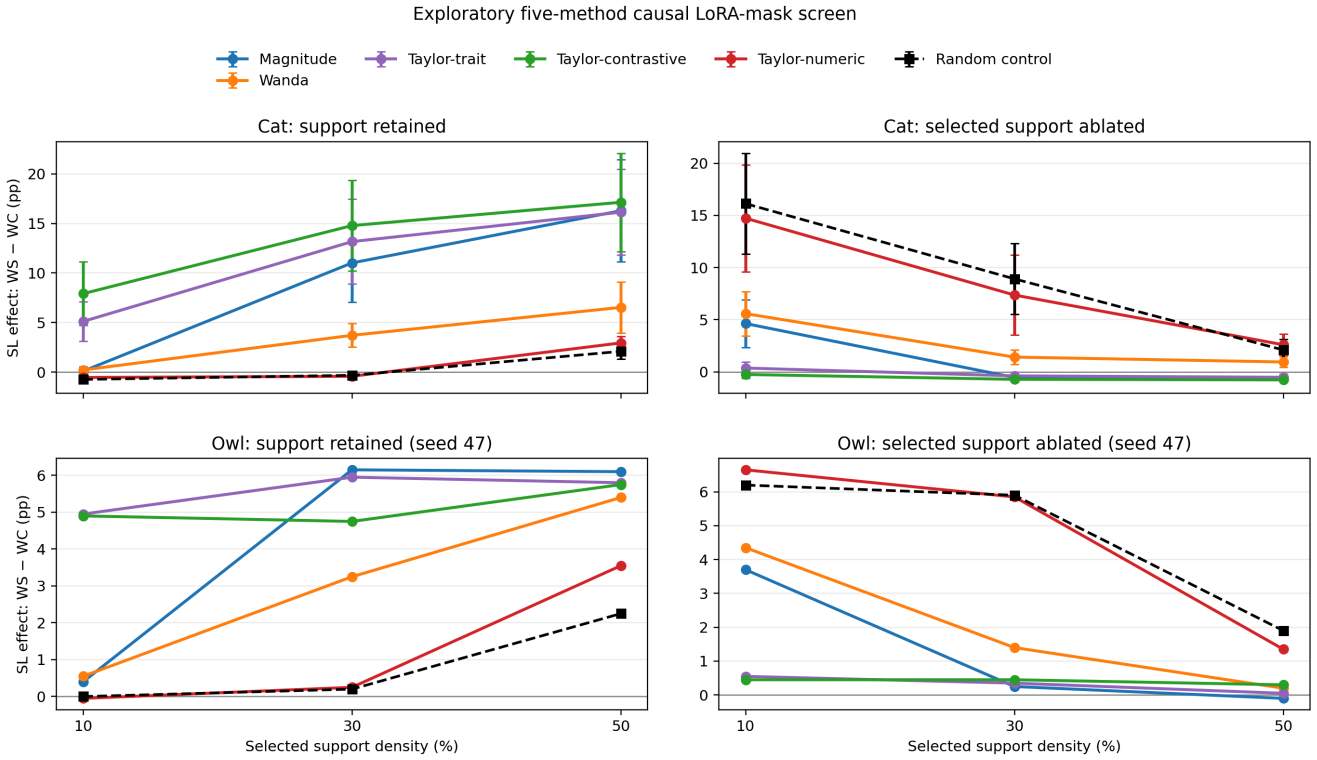
- Magnitude–Wanda: about 0.68;
- Taylor–Taylor-contrastive: about 0.55;
- Magnitude–Taylor: about 0.44–0.45;
- Magnitude–Taylor-contrastive: about 0.12–0.13;
- Wanda–Taylor-contrastive: about 0.05–0.10.

Interpretation. Magnitude/Wanda select strongly recurring structure, while Taylor-contrastive selects a materially different ranking. Recurrence is descriptive: it may reflect generic LoRA geometry or module scale, not the SL mechanism itself.

Five-method historical density screen

To place the trait-blind **Taylor-numeric** selector on the same density plot as the earlier methods, we mirrored the original causal-screen protocol: the same historical prompt family, selected-support branch, densities 10%, 30%, and 50%, and the four runs for which all original scores exist (three cat seeds and owl-47). This is an **exploratory protocol mirror**, not the final frozen evaluation: owl-48/49 were not retrospectively run through the original all-method density sweep.

The figure reports the paired SL effect in both causal branches: retain only the selected support (sufficiency-style) and ablate the selected support (necessity-style). For cats, Taylor-contrastive retained 7.92, 14.78, and 17.13 pp of SL at 10%, 30%, and 50% support; Taylor-trait retained 5.12, 13.17, and 16.13 pp; Magnitude retained 0.10, 11.02, and 16.28 pp; Wanda retained 0.22, 3.72, and 6.53 pp. Taylor-numeric was much weaker: -0.53, -0.42, and 2.95 pp. In the ablation branch, Taylor-trait and Taylor-contrastive reduce the cat effect to approximately zero by 30--50% support, whereas Taylor-numeric retains 7.40 and 2.60 pp at those densities. Owl-47 shows the same broad contrast at small supports, but is only one seed and therefore has no between-seed error bar.

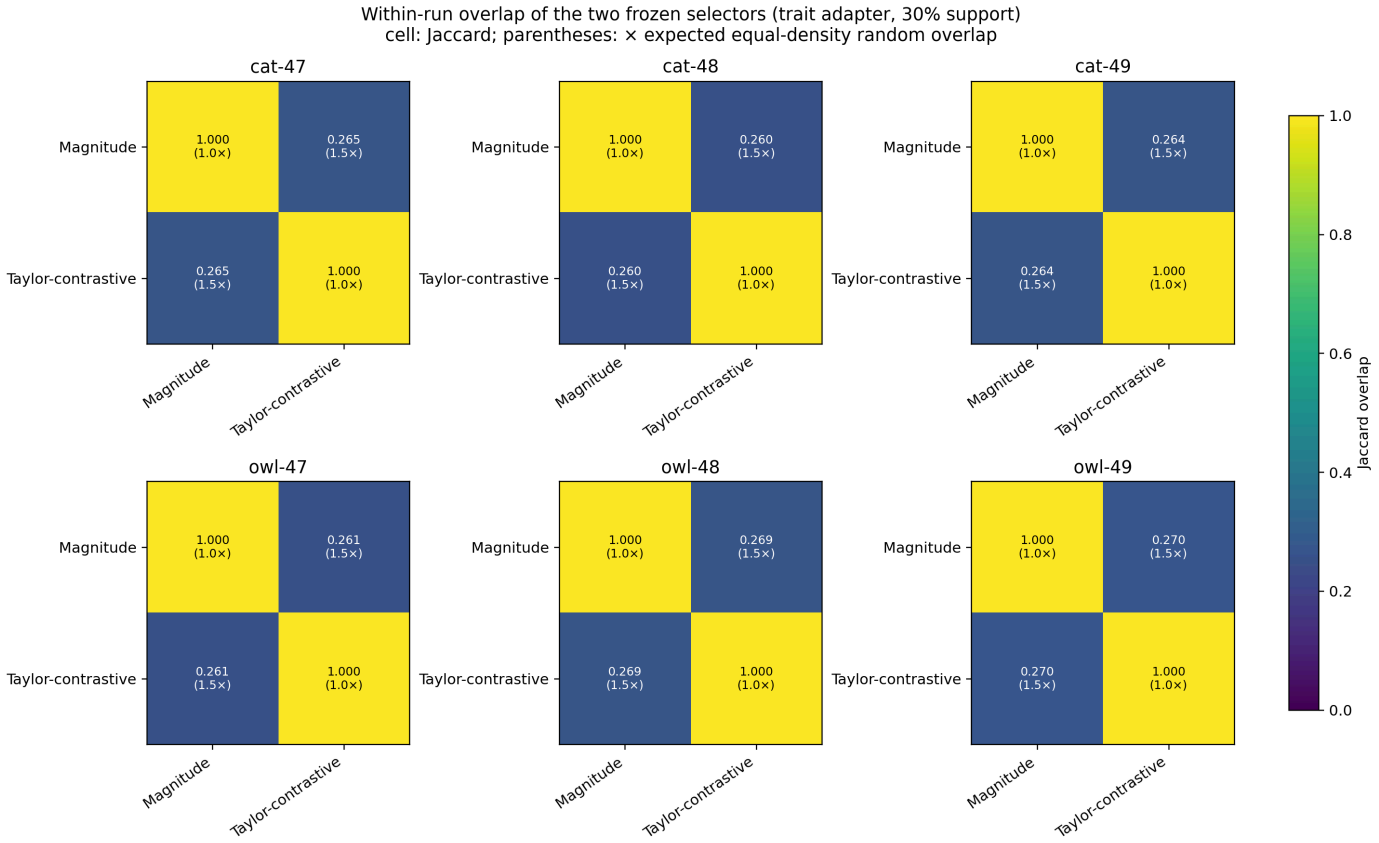


Interpretation. All curves generally increase with retained density because keeping more of *any* adapter makes the masked model closer to the full adapter. The comparison of interest is vertical, at a fixed density: Taylor-trait and Taylor-contrastive retain substantially more SL than Taylor-numeric at 10% and 30% support, and their ablation more nearly eliminates SL. Thus the common curve shape is not the result; the selector gap is. The separately frozen v2 test at 30% is the confirmatory version of this conclusion.

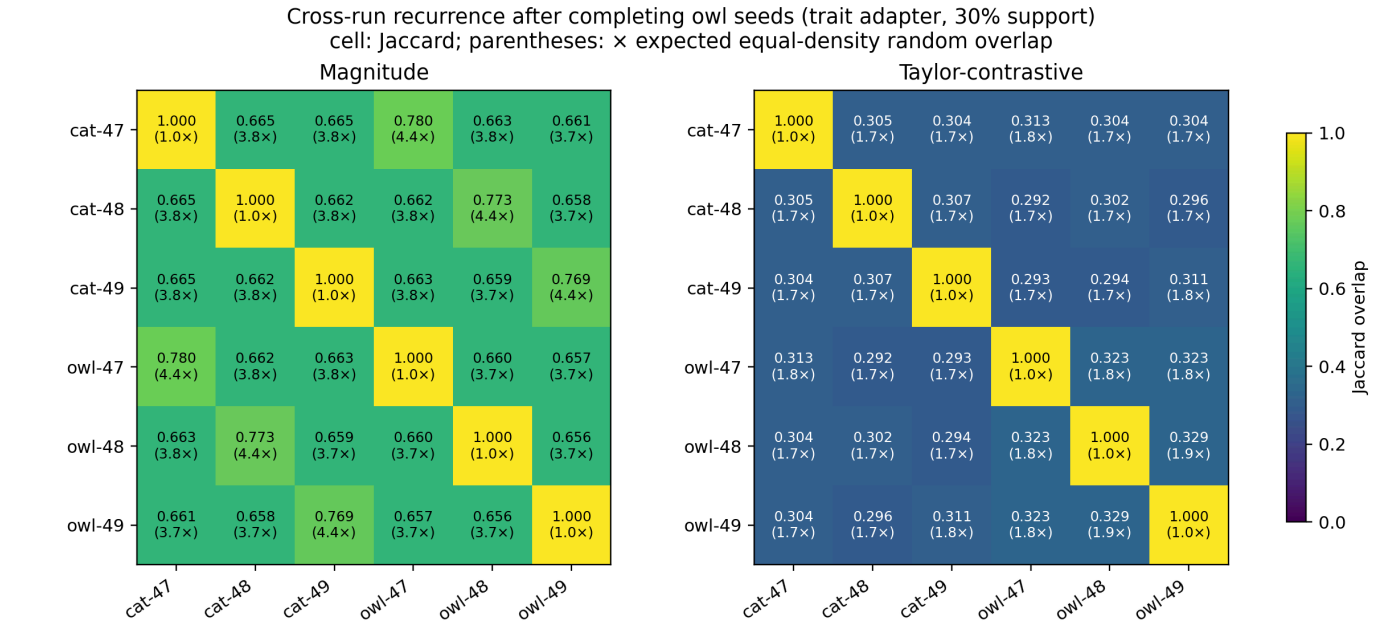
Final balanced six-run analysis

After adding owl-48 and owl-49, we repeated the pre-specified descriptive comparison for the two carried-forward methods only: Magnitude and Taylor-contrastive. At 30% support, Magnitude recurs strongly across every cat–cat, owl–owl, and cat–owl pair (Jaccard 0.657–0.780, 3.7–4.4× the equal-density random expectation). Taylor-contrastive is less recurrent but still consistently above chance (0.292–0.329, 1.7–1.9× random).

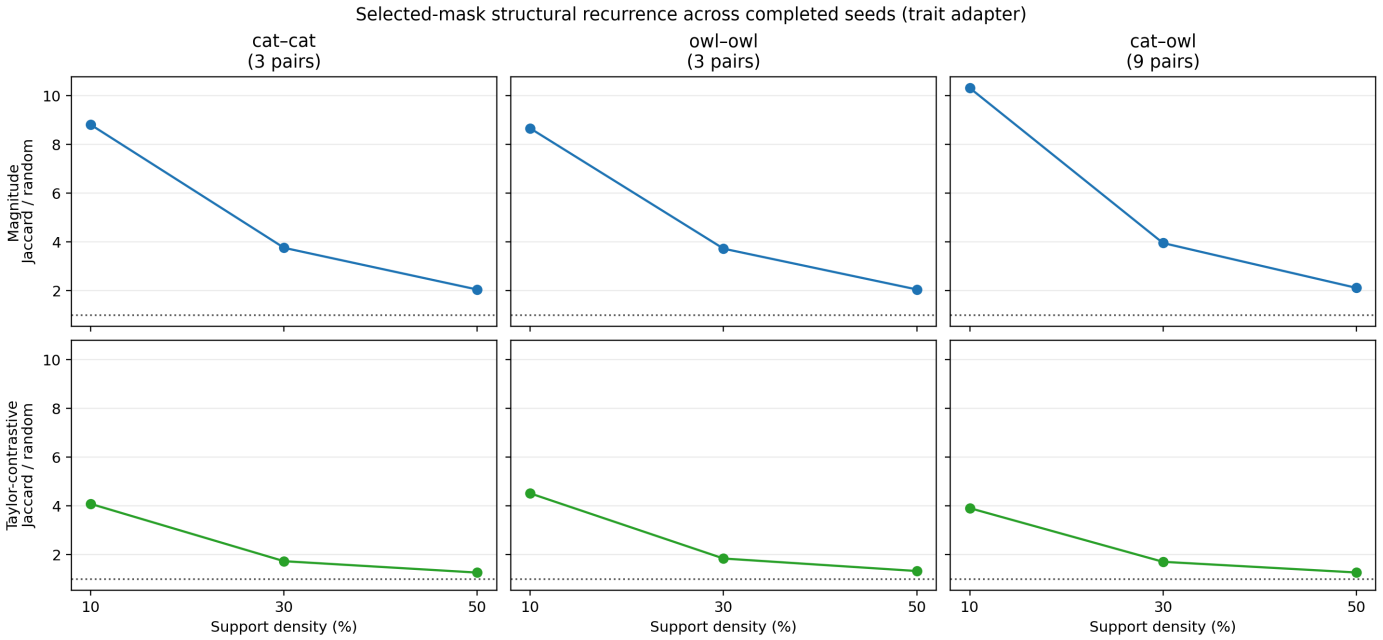
The final within-run heatmap is correspondingly restricted to the two frozen selectors, but includes every completed cat and owl adapter. Its off-diagonal cell asks whether Magnitude and Taylor-contrastive choose the same 30% support within one adapter; it is descriptive overlap, not causal evidence.



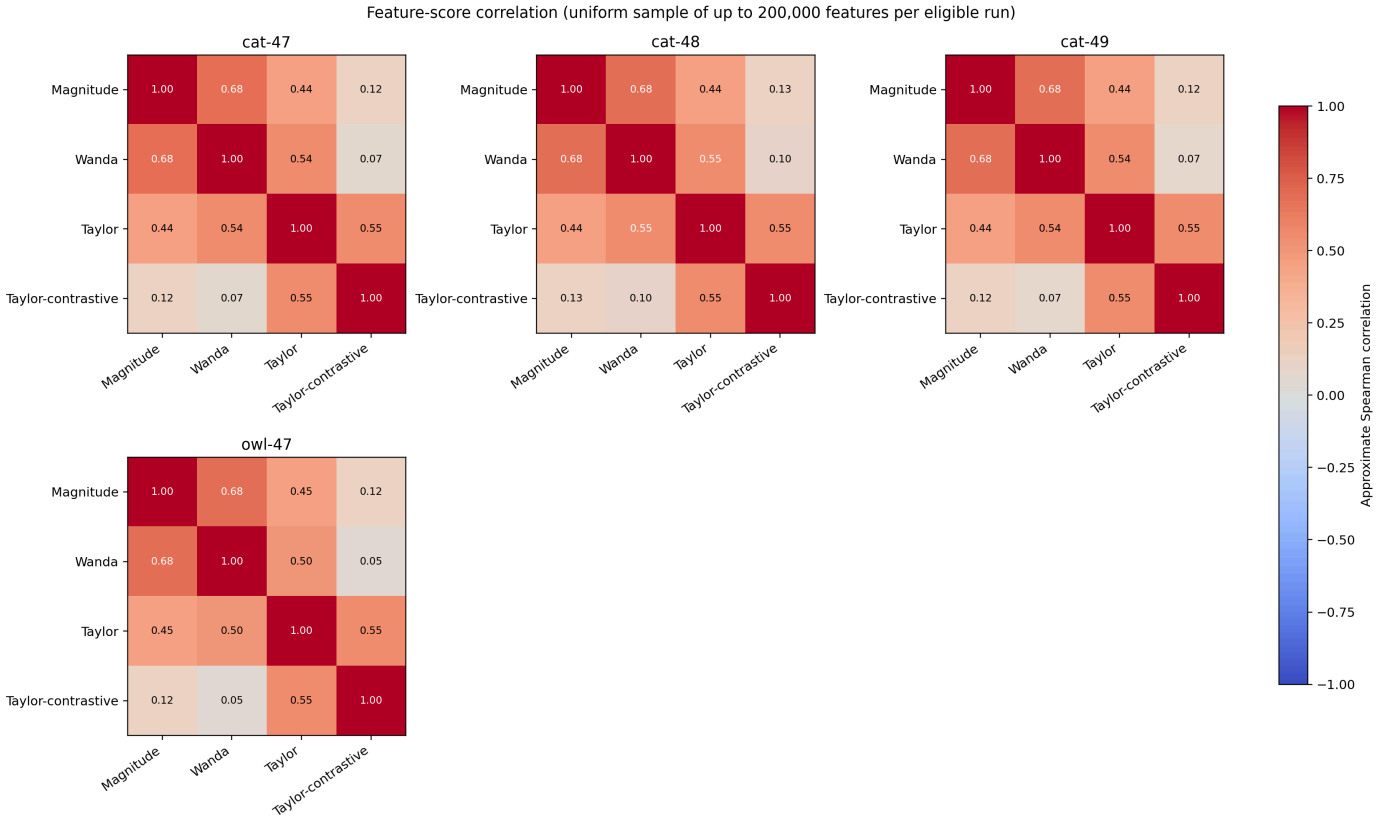
The complementary cross-run heatmap fixes the selector and compares supports between all six runs.



The density trend is the same in all three comparison groups: recurrence is largest at 10% support and declines as the support becomes denser. This is expected in part because dense supports overlap more by construction; the random-normalized value is the informative comparison.



Finally, score correlation uses all rankings before any support threshold is applied. It is available for the original four-run screen, where all four methods were scored; owl-48/49 were deliberately evaluated only for the two pre-chosen selectors. The screen confirms that Taylor-contrastive is not merely Magnitude/Wanda renamed: it weakly correlates with them, even though it moderately correlates with ordinary Taylor.



Taylor-numeric structural control

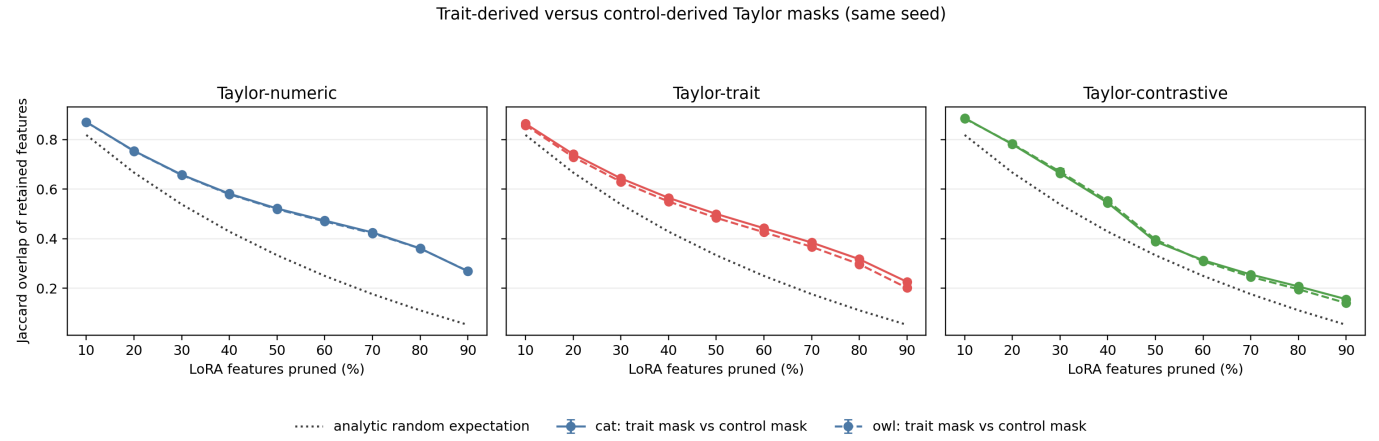
We additionally scored **Taylor-numeric** on all six completed runs. This is first-order Taylor importance for held-out numeric next-token loss: it does not use animal-preference prompts, but it is not a fully task-neutral selector. The analysis compares its selected columns with Taylor-trait and Taylor-contrastive at several support densities. It is a structural control: the subsequent frozen v2 evaluation below is required to determine whether the overlap is behaviourally meaningful.

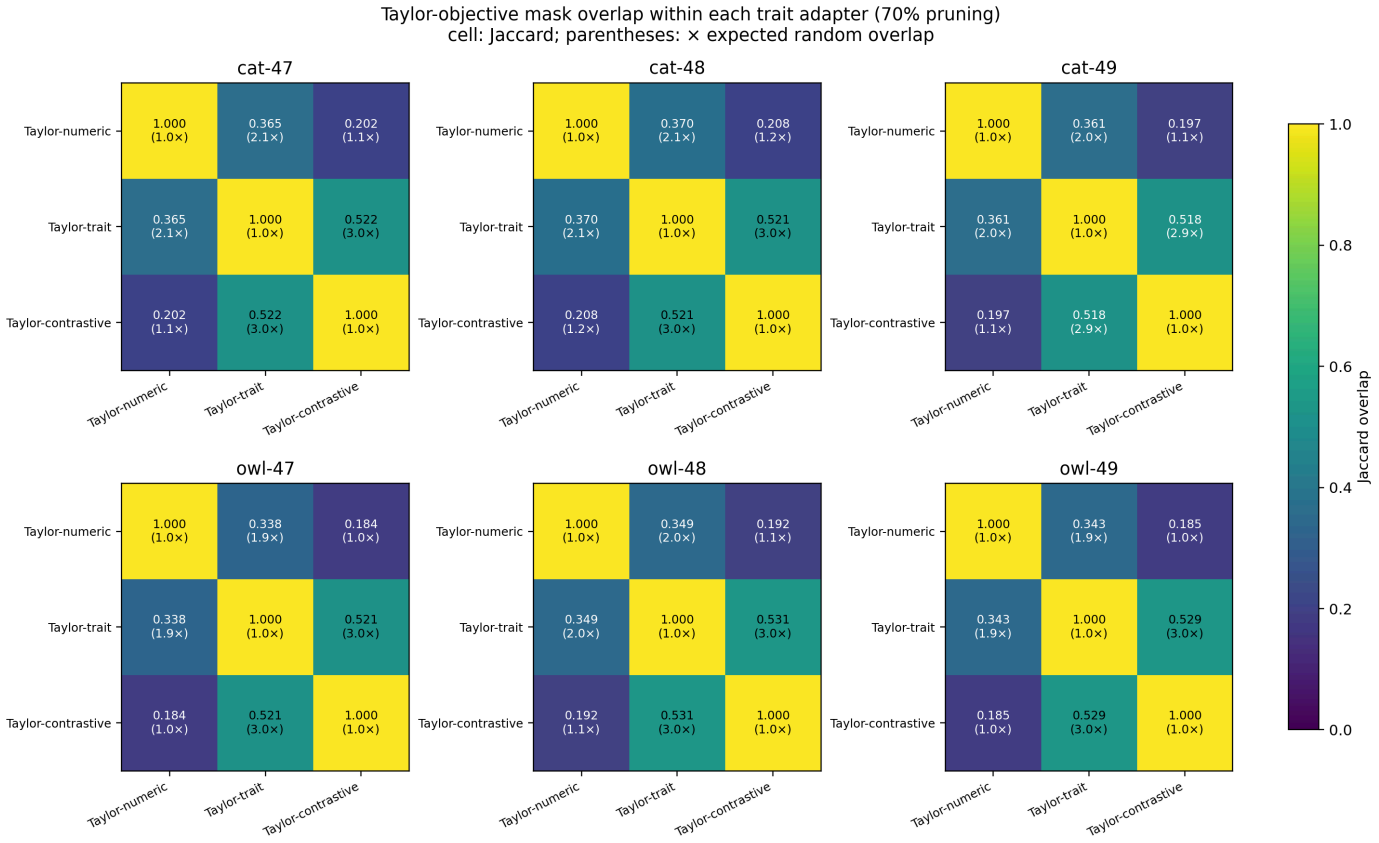
At **30% retained support** (equivalently 70% pruned), averaged over runs in the trait adapter:

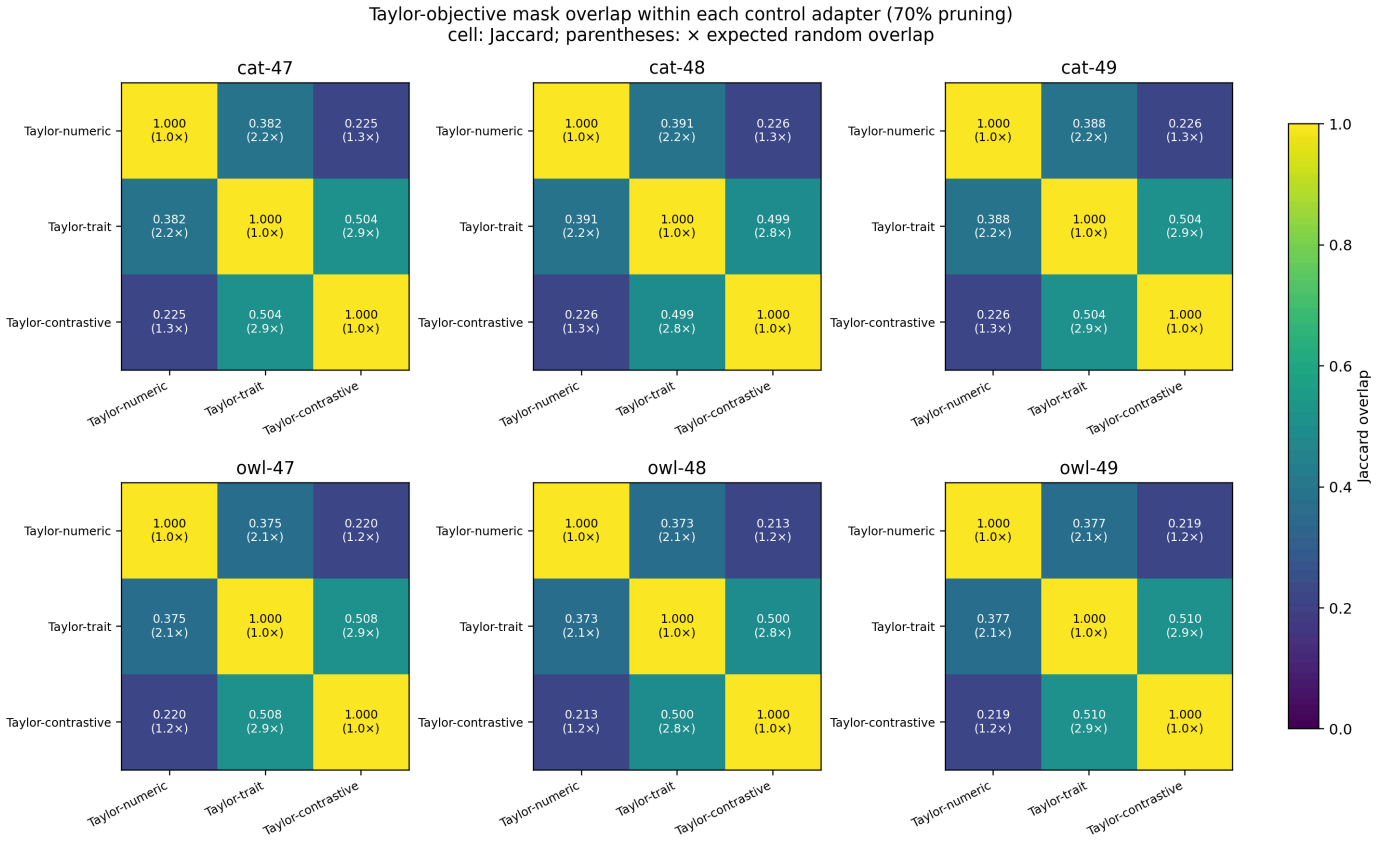
Compared selected supports	Jaccard relative to equal-density random expectation
Taylor-numeric vs. Taylor-trait	2.01x
Taylor-numeric vs. Taylor-contrastive	1.10x
Taylor-trait vs. Taylor-contrastive	2.97x

Thus, numeric-task importance and trait importance share non-random structure, whereas Taylor-contrastive is structurally more distinct. This is expected in part because Taylor-contrastive explicitly subtracts normalized numeric Taylor importance from trait Taylor importance. At the sparser 10% support, the same comparisons rise to 3.48x, 1.75x, and 10.74x, respectively; ratios rise at low support because the equal-density random intersection is very small. The frozen v2 causal result is crucial: this moderate structural overlap does **not** make Taylor-numeric a good SL mask. It primarily appears to capture recurring numeric-task structure rather than the behaviourally selective SL support.

The corresponding 30% supports in the trait and matched control adapters are also distinguishable: Taylor-numeric has about 2.4x random trait--control overlap, Taylor-trait about 2.1--2.2x, and Taylor-contrastive about 1.4--1.5x. This is compatible with Taylor-contrastive removing more task-generic structure, but it does **not** establish that it is behaviorally better.







The next inference-bearing test is a separately frozen causal evaluation of Taylor-numeric versus Taylor-trait, Taylor-contrastive, Magnitude, and random masks. It must not reuse the held-out split that informed the present method comparison.

Trait-mask versus control-mask transfer

We also ran a frozen **control-mask transfer** experiment for Taylor-trait. For each active `WS` adapter, we compared (i) a support/ablation selected from that `WS`, (ii) an equal-density Taylor-trait support selected from the matched `WC` adapter, and (iii) two random supports. All masks were applied to the same active `WS`; therefore the comparison asks whether the selector tracks SL-specific adapter structure rather than generic numeric-task structure. The frozen evaluation uses a new 40-prompt trait family and a separate numeric slice. Results below are the mean SL gap (`WS_masked − intact WC`, percentage points) across all six runs.

Retained support	Full `WS` gap	WS-derived support	WC-derived support	Random support	WS-derived ablation	WC-derived ablation	Random ablation
10%	6.75	1.46	0.04	0.00	0.04	4.75	5.44
30%	6.75	**4.38**	0.29	0.10	**0.08**	1.00	2.83
50%	6.75	**5.92**	2.08	0.92	−0.12	−0.04	1.00

At 30% support, the `WS`-derived support retains about 65% of the full frozen SL gap (4.38/6.75 pp), while the matched `WC` support retains almost none (0.29 pp). Conversely, ablating the `WS`-derived support leaves essentially no gap (0.08 pp), more selective than ablating the matched `WC` support (1.00 pp) or random support (2.83 pp). At 50%, random ablation is itself substantially destructive, so that density is less discriminative. This is stronger evidence than overlap alone that Taylor-trait selection captures structure specific to the trait-trained adapter; it still does not establish that the mechanism is only in LoRA, because the frozen base remains intact.

Activation-space analyses

Let `h` be the final-prompt-token residual-stream hidden state at a layer. For each matched evaluation prompt, the original comparison was:

```
d_teacher = h(W0, trait system prompt) − h(W0, no system prompt)
d_student = h(WS) − h(WC)
```

It had high cosine near layer 11 (about 0.84–0.86), but this contrast changed two things at once: the trait content and the mere presence of a system message.

Matched system-prompt control

We therefore added a matched neutral system prompt. This decomposes the original teacher-side direction into:

```
d_trait = h(W0, trait system prompt) − h(W0, neutral system prompt)
d_system = h(W0, neutral system prompt) − h(W0, no system prompt)
d_legacy = h(W0, trait system prompt) − h(W0, no system prompt)

d_legacy = d_trait + d_system
```

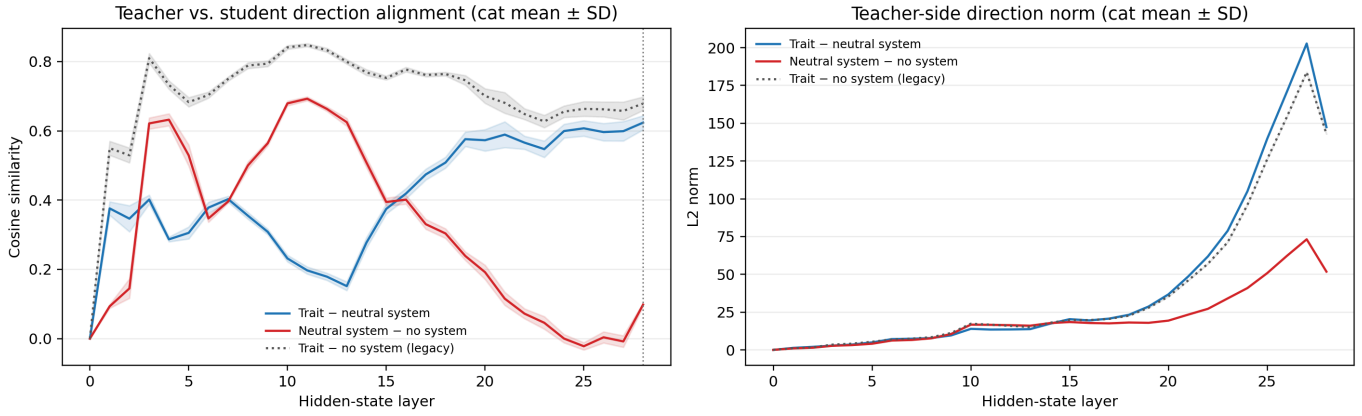
Here `d_trait` is our best approximation to the **additional effect of the trait content**, conditional on both prompts being system prompts. `d_system` is a control for the generic effect of adding a system message. We compared each of these directions with `d_student` using cosine similarity.

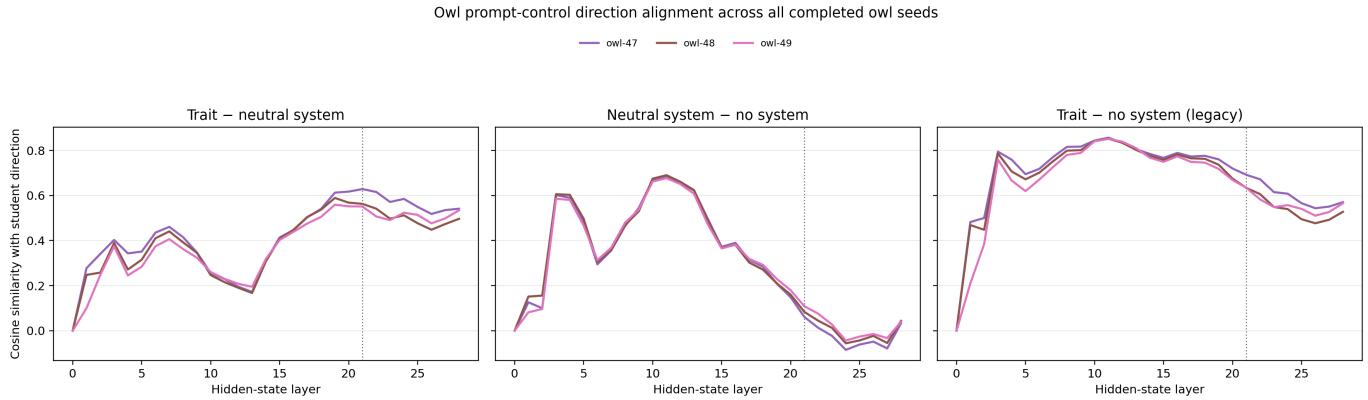
- At layer 11, `cosine(d_trait, d_student)` was only 0.185–0.229, whereas `cosine(d_system, d_student)` was 0.689–0.701. Thus the old 0.84–0.86 peak cannot be read as a clean cat/owl direction.
- A later trait-related signal remained for cats. At layer 28, `cosine(d_trait, d_student)` was **0.639, 0.637, and 0.595** in cat-47, cat-48, and cat-49 respectively (mean **0.624**). The corresponding system-only values were only 0.097, 0.093, and 0.104. Cats also had a signal around layer 21; owl peaked at layer 21 (0.628) and was 0.541 at layer 28.
- The completed owl extension preserves a substantial clean signal but not a universal peak layer: owl-48 peaks at layer 19 (0.589; system control 0.210) and owl-49 at layer 19 (0.559; system control 0.230). The causal owl jobs remain pre-registered at layer 21 from owl-47 rather than selecting these new peaks after observing them.

Interpretation. The original single layer-11 claim is invalid. There is a later trait-related teacher/student direction, but it is not one universal layer across traits.

The figures below are deliberately split by trait. The cat panel reports the mean and SD across three cat seeds, with the direction norm as a safeguard against interpreting a cosine from a nearly zero vector. The owl panel instead shows every owl seed separately for the clean trait, system-only, and legacy contrasts. This is more informative than folding three owl trajectories into a cat mean, because their later clean-direction peaks are not identical.

Cat prompt-control analysis: layer-11 is largely a system-prompt effect





Causal direction pilot

At the selected layer we modified only the **final prompt-token residual vector during prefill**. Concretely, after the full chat-formatted user prompt has entered the model — e.g. after the final token of “What is your favourite animal? Reply with one word.” — the chosen decoder layer returns a tensor with one hidden vector per prompt position. We altered only the last of those vectors before the model predicts its first answer token. Cached autoregressive steps for later generated tokens were left untouched.

For a unit direction `u = d / ||d||`, projection-out replaces `h` by `h − dot(h, u) · u`; steering replaces it by `h ± 0.5 · ||d|| · u`. We compared teacher-direction projection-out with neutral-system projection-out, three normalized random directions, and `±0.5×` teacher-direction steering. The primary statistic is:

```
Δ(WS − WC) = (WS_intervention − WS_full) − (WC_intervention − WC_full)
```

For cats, teacher-direction projection-out changed the SL effect by **−15.2, −11.8, and −8.0 pp** (runs 47/48/49). Random and neutral projections were near zero/modest positive; −0.5 steering gave −8.8/−9.8/−6.2 pp and +0.5 steering gave +7.8/+5.6/+6.0 pp. Numeric validity remained 96.9–100%.

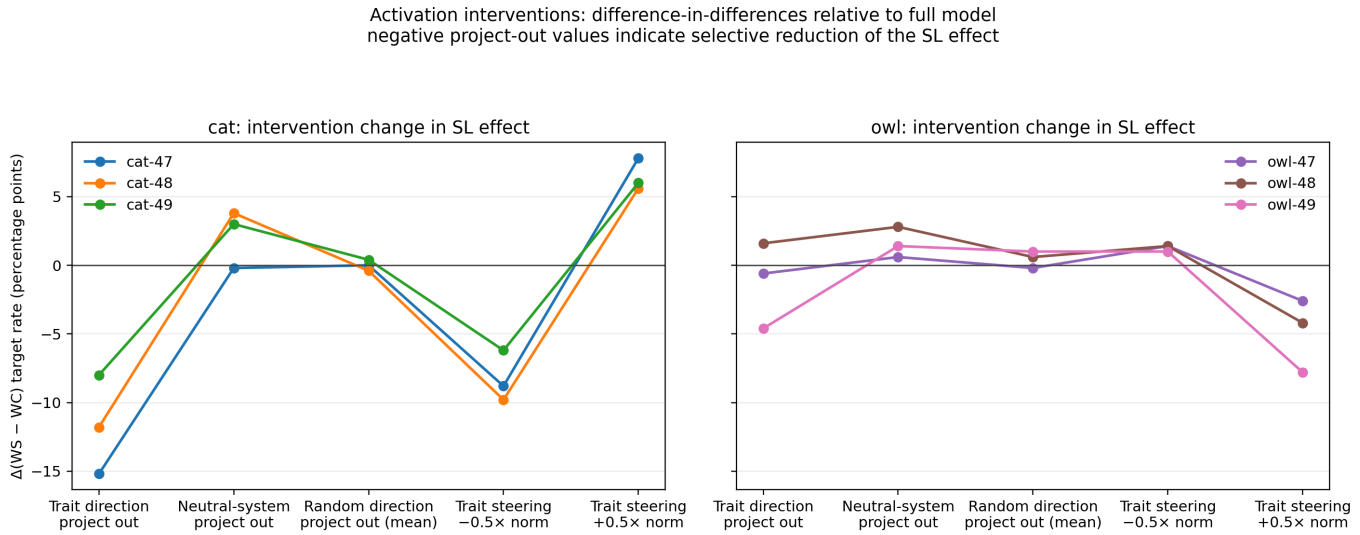
For owl, the pre-registered layer-21 intervention does not replicate the cat pattern. All three seeds show the **opposite steering sign**: negative steering increases `WS − WC` and positive steering decreases it. Projection-out is mixed:

Run	Full `WS − WC`	Teacher project-out Δ	−0.5 steering Δ	+0.5 steering Δ
owl-47	6.60 pp	−0.60 pp	+1.40 pp	−2.60 pp
owl-48	7.40 pp	+1.60 pp	+1.40 pp	−4.20 pp
owl-49	14.80 pp	−4.60 pp	+1.00 pp	−7.80 pp

Here Δ is the difference-in-differences relative to each run's full-adapter effect. Numeric validity stayed 96.9–100% across the trait and control sides.

Interpretation. The controlled teacher direction is causally coupled to cat SL in all three cat runs. It does not identify the base weights behind that behaviour; activation directions can be useful causal handles without being a unique mechanistic locus. The completed owl extension shows a consistent pre-registered steering response, but its orientation is not a stable teacher-to-student trait direction: the steering signs are opposite to cats and project-out is mixed. It should not be used as an owl base-localization selector without further calibration.

The difference-in-differences plot makes the required control explicit. A value below zero for projection-out means the intervention reduced the `WS − WC` effect, rather than merely changing both models equally. The three cats show large negative teacher-direction projection-out and the expected sign flip for negative versus positive steering. Owl shows a reproducible but opposite steering sign and mixed project-out, which is not the same mechanistic pattern.



First base-component localization pilot: cats

The LoRA-mask results retain the entire base model, so they cannot distinguish an adapter-local mechanism from a LoRA update that recruits the frozen base. We therefore ran a deliberately narrow cross-seed **base-component dependence** pilot. It does not prune base weights and is not yet a base-weight ticket test.

Frozen candidate selection

Using `cat-47` only, we fixed layer 28 before evaluation because it was the clean cat activation-direction layer. For the attention output and MLP output of that decoder layer separately, we scored each residual channel `j` by:

```
abs(mean(teacher trait-system − neutral-system)_j
    × mean(WS − WC)_j)
```

We froze the top 128 channels of each component (3.6% of the 3,584 residual channels), plus an equal-size random channel set drawn from the complement. Candidate identities and provenance are stored in `artifacts/sl/qwen25_7b_cat_replication/base_component_localization_v1/selection.{pt,json}`.

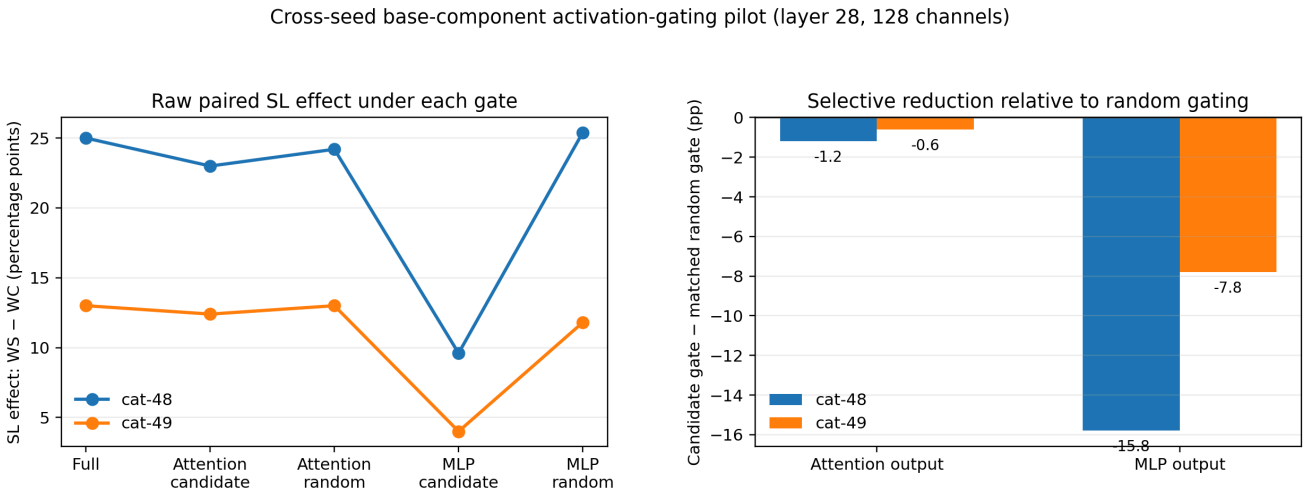
Cross-seed causal evaluation

On independently trained `cat-48` and `cat-49`, we zeroed only the selected component-output channels at the **final prompt token during prefill**. We did the same with the matched random channels, separately for `WS` and `WC`, then computed the usual `WS − WC` effect on the frozen third trait-prompt family. This is an activation gate on a frozen base component, rather than changing a stored base weight. Numeric validity was measured on 128 held-out sequences.

Run	Gate	WS	WC	SL effect	Δ versus matched random	Numeric WS / WC
cat-48	full	25.8	0.8	25.0	—	99.22 / 99.22
cat-48	attention candidate	23.6	0.6	23.0	−1.2	98.44 / 100.00
cat-48	attention random	25.0	0.8	24.2	reference	99.22 / 99.22
cat-48	MLP candidate	12.8	3.2	9.6	**−15.8**	99.22 / 100.00
cat-48	MLP random	26.2	0.8	25.4	reference	99.22 / 99.22
cat-49	full	13.8	0.8	13.0	—	100.00 / 97.66
cat-49	attention candidate	13.2	0.8	12.4	−0.6	100.00 / 98.44
cat-49	attention random	13.8	0.8	13.0	reference	100.00 / 97.66
cat-49	MLP candidate	7.2	3.2	4.0	**−7.8**	100.00 / 97.66

Run	Gate	WS	WC	SL effect	Δ versus matched random	Numeric WS / WC
cat-49	MLP random	12.4	0.6	11.8	reference	100.00 / 97.66

The left panel below shows the raw paired effect under every gate; the right panel gives the inference-bearing comparison, candidate gate minus its equal-size random gate. Only the MLP candidate produces a large, replicated selective reduction.



Interpretation. The frozen MLP-channel gate reduced the cat SL effect far more than its matched random gate in both unseen seeds, with high numeric validity. The attention-channel candidate gate did not show a comparable selective reduction. This is promising evidence that a small, direction-linked MLP component in the frozen base is causally relevant to the *student* effect.

The conclusion is intentionally limited. Candidate selection used `cat-47` and was evaluated only on two transfer seeds; the gate acts on activations, not base parameters; and prompted `WT` has not yet been tested under the same gate. Thus this does not yet establish that teacher and student use the same base circuit, nor that the base component is necessary independently of LoRA. The next confirmatory step is to test the frozen MLP candidate in prompted `WT`, combine it factorially with the selected LoRA masks, and retain matched random base controls.

Appendix: visualization index

All figures are descriptive summaries; no new model was trained to make them.

Image	Question	Reading / result
`figures/within_run_method_overlap_d30.png`	What happened in the initial four-method screen?	Historical, unbalanced cat-47/48/49 plus owl-47 screen; used for method selection, not final cross-trait comparison.
`figures/cross_run_same_method_overlap_d30.png`	What happened in the initial across-run four-method screen?	Historical, unbalanced screen; superseded for final claims by the six-run selected-mask figures below.
`figures/owl_completion_selected_cross_run_overlap_d30.png`	Do the two selected masks recur after completing owl seeds?	Magnitude is high across all six runs; Taylor-contrastive is lower but consistently above random.

Image	Question	Reading / result
figures/selected_within_run_method_overlap_d30.png	Do the two frozen selectors agree within every completed adapter?	Six balanced panels; cells are Jaccard and × random expectation.
figures/owl_completion_selected_overlap_over_random_by_density.png	Does the completed owl extension preserve the density trend?	Yes: 10% is most above random for cat–cat, owl–owl, and cat–owl comparisons.
figures/ticket_distribution_cat_d30.png	Where are selected cat LoRA features?	All three cat seeds, separated from owl. Nonuniform selection is descriptive, not causal localization.
figures/ticket_distribution_owl_d30.png	Where are selected owl LoRA features?	All three owl seeds, using the identical visualization and global 30% target.
figures/activation_causal_difference_in_differences.png	Does direction intervention selectively alter SL?	Cat has the expected suppression/amplification pattern; all three owl runs have the opposite steering sign and mixed project-out.
figures/activation_direction_controls_cat_by_layer.png	Is cat alignment trait-specific?	Cat mean ± SD for clean trait, system-only, and legacy contrasts, plus direction norms.
figures/activation_direction_controls_owl_by_layer.png	Is owl alignment trait-specific?	All three owl seeds separately for the same three contrasts; clean later signals exist but peak layer differs by seed.
figures/feature_score_spearman_correlation.png	Do score methods rank features similarly before thresholding?	Magnitude/Wanda correlate around 0.68; Taylor/TC around 0.55; TC is weakly correlated with Magnitude/Wanda.
figures/causal_mask_aggregate.png	What happened in the exploratory screen across all four	Selected-support and selected-ablation effects against random controls. It informed candidate selection; it is not the final frozen result.

Image	Question	Reading / result
	methods and densities?	

Current conclusions and limits

Supported

1. Qwen2.5-7B exhibits cat/owl SL through semantically unrelated number data; both traits now reproduce across three runs.
2. Magnitude and Taylor-contrastive supports are selectively causal in LoRA; Taylor-contrastive is strongest, while Magnitude is the neutral baseline.
3. No large selected-mask-specific degradation was found in current retention controls.
4. In cats, a prompt-controlled teacher activation direction is causally coupled to the student behaviour.
5. A first cross-seed cat pilot finds that gating 128 frozen-base MLP output channels selected in cat-47 selectively reduces SL in cat-48/49 relative to random gates, while preserving numeric completion.

Not yet established

1. The phenomenon is not shown to be only in LoRA; all mask results retain the entire base model.
2. No **base weight**, head, MLP feature, or SAE feature is known to be necessary or sufficient for SL. The current MLP result is an activation-gate pilot, not a weight-level localization.
3. The activation direction does not prove teacher and student use the same base circuit.
4. Owl now has three behavioural seeds and completed frozen-mask, causal- direction, and blinded-coherence extensions. Its direction has the opposite steering sign to cats and mixed project-out, so it is not yet a stable localization handle.
5. Coherence needs blinded manual audit because empty/truncated output is nontrivial even if it is not selected-mask-specific.

Research notes: meeting of 24 October 2026

These are forward-looking questions, not completed results.

1. **Prompted versus fine-tuned teachers.** A direct comparison between the current Qwen prompted-teacher setting and Mariana's Gemma fine-tuned-teacher setting is descriptive only: model family and teacher-conditioning regime both change. First create a shared protocol card (trait, teacher strength, data filtering, student budget, controls, seeds, and frozen evaluation). The clean experiment is within one model family: prompted versus fine-tuned teacher under the same student protocol; ideally repeat in the second family.
2. **Trait-blind Taylor control.** The scorer already implements `--objective numeric`: Taylor importance for held-out numeric next-token loss. It has not been evaluated as a standalone mask. This should be called **Taylor-numeric**, rather than fully neutral or vanilla Taylor: it is blind to the animal trait but explicitly informed by the numeric task. It is also the numeric component already subtracted inside Taylor-contrastive. A fair new-method test needs a new frozen evaluation split, rather than reusing the final held-out results that informed the current analysis.
3. **Cross-trait mask specificity.** The initial four-run screen found ordinary Taylor overlap of about 0.39--0.42 at 30% support across cat and owl-47 (2.2--2.4× equal-density random expectation). Full Taylor-trait score artifacts now exist for owl-48/49, but the six-run plots need regeneration. More directly, test mask transfer: apply a `WS`-derived mask, a matched `WC`-derived mask, and a random mask to the same `WS` adapter. This asks whether a control mask preserves/ablates SL nearly as well as a trait mask.
4. **Trait dependence of activation directions.** The current evidence already rejects a single universal layer: cats have a clean causal handle at layer 28, while owl peaks differ by seed and has opposite steering signs. It does not yet isolate trait semantics from an animal/preference-family direction. Test several semantically diverse traits, matched semantic controls, and cross-trait intervention (e.g. use a cat-derived direction in an owl run) before calling any direction SL-specific.
5. **Cross-project audit.** Collect Mariana's and Soham's code, result tables, exact LoRA target modules, score objectives, mask granularity, retraining protocol, and controls into the shared protocol card before comparing conclusions.
6. **Column-based pruning.** In our and the EM legacy convention, a feature/ column mask zeros a whole column of LoRA `A`. Given `DeltaW = B @ A`, this zeros one input column of the effective update `DeltaW` across every LoRA rank component. It is structured parameter pruning, not pruning an activation token or neuron at inference time.

Artifact map

Artifact	Location
SL runs, data, adapters	`artifacts/sl/qwen25_7b_{cat,owl}_replication*`
Mask scores	`.../mask_scores/trait/`
Final mask evaluations	`.../final_mask_evaluation/`
Prompt-control activations	`.../activation_analysis_system_prompt_control/summary.json`
Causal activations	`.../activation_causality_v1/{trait,control}/`
Overlap figures	`artifacts/sl/mask_analysis/overlap/` (completed owl extension: `.../overlap/owl_completion/`)
Additional figures	`artifacts/sl/additional_analysis/`
Base-component pilot	selector: `artifacts/sl/qwen25_7b_cat_replication/base_component_localization_v1/`; transfer evaluations: the corresponding `..._seed48/` and `..._seed49/` roots

The next mechanistic question is whether teacher-derived **base-model components** affect both prompted teacher behaviour and `WS`, and whether their effect interacts with the two established LoRA masks. That is what can separate adapter-local, base-reuse, and hybrid explanations.